CareSwitch - Source code and dependency inventory

Complete first-party deployed app source, including HTML, CSS, JavaScript modules and workers.
The full runnable source ZIP and GitHub repository include unmodified third-party JS/WASM,
the model and their licenses. Binary assets and upstream runtime source are not printed here.
Non-ASCII source characters appear as reversible Unicode escapes for font compatibility.
Use the ZIP/repository for the original, directly runnable UTF-8 files.
Source: https://github.com/syedkaifali96/careswitch

Dependencies: MediaPipe tasks-vision 1.1.0 (Apache-2.0); Face Landmarker model version 1;
eSpeak NG npm 1.0.2 (GPL-3.0-or-later). Attribution/source references accompany assets.
AI tools assisted planning, coding, debugging, testing and documentation.
This is a student prototype, not a clinically validated assistive medical device.

Files / SHA-256 (first-party source):
index.html
997f81cc9219a9a1dfc04b1f1e5204fbaa343c07ae034bc2dfd06a64cbc6b5f2
styles.css
9c0c3338ec564c05d77bae452e174c38197ca88f3886e61101da2a35ac122274
app.mjs
bc64c3d281e2e0150db2f3c08b09ad28f436cc2ed0b60f8e1bdf3359d1657474
core.mjs
cafee77d817daa94b61064e66bdeb58680d309be738797267158313ae7376dd2
pointer.mjs
36f8df1b981fc9b87178c02d13f845ad26c3fe0ec0db897a0cc49d0431e2183e
camera.mjs
dabd6484c2d3eceb3033027382c18c08fd6266bc98d0a492edddcf8e5392f4a2
vision-worker.js
56aac1c821156a8b73b45434052f575649143861640f4618ed278511a6949754
speech.mjs
b0efe657bdd3c755b44542e7b76b198ed1a464322d8fc7f48650a617b701f818
speech-worker.mjs
df7e17633ab743bf8e81b15ab7ac17f3ae5ee0f2abf63ae85f867e15506b3cae

dist/index.html

```
<!doctype html>
<html lang="en">
<head><meta charset="utf-8"><meta name="viewport" content="width=device-width,initial-scale=1"><meta name="description" content="Communicate with head movement using your webcam. CareSwitch is an assistive communication prototype with on-device AI and message confirmation."><title>CareSwitch \xb7 Your movement. Your message.</title><link rel="icon" href="data:image/svg+xml,%3Csvg xmlns='http://www.w3.org/2000/svg' viewBox='0 0 32 32'%3E%3Crect width='32' height='32' rx='9' fill='%232541b2'/%3E%3Cpath d='M8 9h16v11H15l-5 4v-4H8z' fill='white'/%3E%3Ccircle cx='12' cy='14' r='1' fill='%232541b2'/%3E%3Ccircle cx='16' cy='14' r='1' fill='%232541b2'/%3E%3Ccircle cx='20' cy='14' r='1' fill='%232541b2'/%3E%3C/svg%3E"><link rel="stylesheet" href="styles.css"><script type="module" src="app.mjs"></script></head>
<body>
<a class="skip-link" href="#message-grid">Skip to messages</a>
<header><div class="brand"><span class="brand-mark" aria-hidden="true"><svg viewBox="0 0 24 24"><path d="M5 4h14a2 2 0 0 1 2 2v10a2 2 0 0 1-2 2h-8l-6 4v-4a2 2 0 0 1-2-2V6a2 2 0 0 1 2-2Z"/><path d="M7 10h10M7 14h6"/></svg></span>Care<span>Switch</span></div><span class="header-note">Hands-free communication</span><div class="header-actions"><span class="prototype">Prototype</span><button id="settings-button" class="quiet settings-button"><svg aria-hidden="true" viewBox="0 0 24 24"><path d="M4 7h16M4 17h16"/><circle cx="9" cy="7" r="3"/><circle cx="15" cy="17" r="3"/></svg>Settings</button></div></header>
<main>
  <div class="intro"><div><p class="eyebrow">YOUR MOVEMENT. YOUR MESSAGE.</p><h1>Say it your way.</h1><p>Choose a message. Confirm it. Let your voice be heard.</p></div><div class="intro-tools"><a class="camera-jump secondary" href="#controls">Camera setup</a><div class="language"><label for="language">Message language</label><select id="language"><option value="en">English</option><option value="ur">\u0627\u0631\u062f\u0648 \xb7 Urdu</option><option value="ar">\u0627\u0644\u0639\u0631\u0628\u064a\u0629 \xb7 Arabic</option><option value="hi">\u0939\u093f\u0928\u094d\u0926\u0940 \xb7 Hindi</option><option value="es">Espa\xf1ol \xb7 Spanish</option><option value="fr">Fran\xe7ais \xb7 French</option><option value="de">Deutsch \xb7 German</option><option value="zh">\u4e2d\u6587 \xb7 Chinese</option></select></div></div></div>
  <div class="layout">
    <aside class="control-panel" id="controls" aria-labelledby="controls-heading">
      <div class="control-title"><h2 id="controls-heading">Your controls</h2><span class="local-tag">On-device AI</span></div>
      <div class="camera-preview"><video id="video" muted playsinline autoplay aria-label="Your camera preview"></video><div id="camera-placeholder"><span aria-hidden="true"><svg viewBox="0 0 48 48"><rect x="7" y="13" width="26" height="24" rx="5"/><path d="m33 21 9-5v19l-9-5M20 7v6M14 7h12"/></svg></span><strong>Ready when you are</strong><p>Your camera is off.</p></div><div class="camera-state" id="camera-state">Not recording</div></div>
      <div class="tracking-status" id="tracking-status" role="status" aria-live="polite">Start with the camera, or try the board without it.</div>
      <button class="primary" id="start-camera">Enable camera</button>
      <div class="camera-actions"><button class="secondary" id="calibrate" aria-label="Set my movement range" disabled><span class="wide-label">Set my movement range</span><span class="compact-label">Set range</span></button><button class="quiet" id="stop-camera" disabled>Turn camera off</button></div>
      <div class="control-divider"></div>
      <button class="secondary full" id="practice">Try without camera</button><p class="hint">Tap or click a message, or practice holding with a mouse. No AI tracking in practice mode.</p>
      <details class="instructions"><summary>How head control works</summary><ol><li>A helper can enable the camera and start setup.</li><li>Follow five short prompts using only comfortable movements.</li><li>Move your head to guide the pointer. Hold over a message to select it.</li><li>Move to Confirm or Cancel, then hold again.</li></ol><p>If you move your chair or camera, set your movement range again.</p></details>
      <div class="privacy"><svg aria-hidden="true" viewBox="0 0 24 24"><path d="m12 3 8 3v6c0 5-8 9-8 9s-8-4-8-9V6l8-3Z"/><path d="m8 12 3 3 5-6"/></svg><div><strong>Your camera stays private.</strong><p>Images are processed on this device. No video upload or recording.</p></div></div><a class="board-jump quiet" href="#board-heading">Back to messages</a>
    </aside>
    <section class="communication" aria-labelledby="board-heading">
      <div class="board-heading"><div><p class="eyebrow">COMMUNICATION BOARD</p><h2 id="board-heading">What would you like to say?</h2></div><button id="pause" class="pause" data-target="pause" aria-label="Pause head control" disabled><span class="wide-label">Pause head control</span><span class="compact-label">Pause</span></button></div>
      <div class="board-state"><span id="mode-label">Tap or click a message to begin.</span><span id="hold-label">Hold to select \xb7 1.5 seconds</span></div>
      <div class="tracking-surface" id="surface">
        <div class="message-grid" id="message-grid" tabindex="-1"></div>
        <section id="confirmation" class="confirmation" aria-labelledby="confirm-heading" hidden><p class="eyebrow">YOU CHOOSE WHAT\u2019S SAID</p><h3 id="confirm-heading">Is this your
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dist/index.html (continued)

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+       message?</h3><p id="selected-message"></p><p id="voice-note" class="voice-note"></p><div
+     class="confirm-actions"><button class="confirm-button" id="confirm" data-
+     target="confirm">Confirm message</button><button class="cancel-button" id="cancel" data-
+     target="cancel">Cancel</button></div></section>
27            <section class="message-output" aria-labelledby="output-heading"><div class="output-
+     label"><h3 id="output-heading">Your confirmed message</h3><div class="output-actions"><button
+     id="repeat" class="quiet" disabled>Read again</button><button id="clear" class="quiet"
+     disabled>Clear</button></div></div><p id="output" class="empty" role="status" aria-
+     live="polite">Your message will appear here.</p><span id="output-status">Nothing is spoken
+     until you confirm.</span></section>
28            <div id="pointer" class="head-pointer" aria-hidden="true" hidden><svg viewBox="0 0 44
+     44"><circle class="ring-bg" cx="22" cy="22" r="18"></circle><circle id="pointer-ring" cx="22"
+     cy="22" r="18"></circle></svg><span></span></div>
29          </div>
30          <div class="below-board"><span id="selection-status" role="status">No message
+     selected.</span><button id="edit-phrases" class="quiet">Customize messages</button></div>
31        </section>
32      </div>
33      <footer><span>CareSwitch \xb7 AI \xd7 Healthcare</span><span>Communication prototype \xb7
+     not an emergency alert or clinically validated device</span></footer>
34    </main>
35    <dialog id="settings-dialog"><form method="dialog" class="dialog-heading"><h2>Make it
+     comfortable</h2><button class="close" aria-label="Close settings">\xd7</button></form><p>Use
+     movements that feel comfortable. A helper can adjust these controls.</p><label class="slider-
+     label" for="dwell">Hold time <output id="dwell-output">1.5 seconds</output></label><input
+     id="dwell" type="range" min="1" max="4" step="0.5" value="1.5"><label class="slider-label"
+     for="sensitivity">Movement sensitivity <output id="sensitivity-
+     output">0.8\xd7</output></label><input id="sensitivity" type="range" min="0.6" max="1.6"
+     step="0.1" value="0.8"><label class="slider-label" for="smoothing">Pointer smoothing <output
+     id="smoothing-output">Strong</output></label><select id="smoothing"><option
+     value="0.45">Light</option><option value="0.25">Medium</option><option value="0.12"
+     selected>Strong</option></select><label class="check"><input type="checkbox"
+     id="reverse-x">Reverse left / right direction</label><p class="hint">Small movements are
+     filtered to help you hold a message. Lower sensitivity slows movement; Strong smoothing makes
+     the pointer steadier. If tracking stops, selection stops too. Escape cancels a draft message.
+     Space pauses active head control when you\u2019re not editing text.</p><label class="slider-
+     label" for="voice-mode">Voice</label><select id="voice-mode"><option value="auto">Automatic
+     \xb7 device voice first</option><option value="basic">Free built-in voice</option></select><p
+     id="voice-settings" class="hint"></p><button class="primary" id="settings-
+     done">Done</button></dialog>
36    <dialog id="calibration-dialog"><h2 id="calibration-title">Let\u2019s set your movement
+     range.</h2><p id="calibration-text">Sit comfortably with your face visible.</p><div
+     class="calibration-diagram"><span class="cal-dot" id="cal-dot"></span><span class="cal-center"
+      aria-hidden="true">+</span></div><div class="cal-progress"><span id="cal-
+     progress"></span></div><p id="calibration-step" class="hint">Step 1 of 5</p><button
+     class="secondary" id="cancel-calibration">Cancel setup</button></dialog>
37    <dialog id="phrases-dialog"><form method="dialog" class="dialog-heading"><h2>Your
+     messages</h2><button class="close" aria-label="Close message
+     editor">\xd7</button></form><p>Edit the six messages in the selected language. Use any writing
+      system. Messages stay in this tab and reset when you reload.</p><div id="phrase-
+     fields"></div><button class="primary" id="save-phrases">Use these messages</button><p
+     id="phrase-error" class="hint" role="status"></p></dialog>
38    </body></html>
```

dist/styles.css

```
:root {
  --ink: #152642; --muted: #52647d; --blue: #234bc5; --blue-dark: #193a9b;
  --blue-soft: #edf3ff; --line: #dce4ef; --bg: #f3f6fb; --yellow: #ffcd46;
  --navy: #142540; --radius: 20px;
  color-scheme: light;
}
* { box-sizing: border-box; }
html { scroll-padding-top: 24px; }
body { margin: 0; background: var(--bg); color: var(--ink); font-family: 'Segoe UI', Inter,
Arial, sans-serif; font-size: 16px; line-height: 1.5; }
button, input, select { font: inherit; }
button, select, summary { -webkit-tap-highlight-color: transparent; }
button { cursor: pointer; border-radius: 12px; transition: background .16s, border-color .16s,
 box-shadow .16s; touch-action: manipulation; }
button:disabled { opacity: .48; cursor: not-allowed; }
button:focus-visible, input:focus-visible, select:focus-visible, summary:focus-visible,
a:focus-visible { outline: 3px solid #2b64da; outline-offset: 4px; }
[hidden] { display: none !important; }
svg { fill: none; stroke: currentColor; stroke-width: 1.8; stroke-linecap: round; stroke-
linejoin: round; }
.skip-link { position: fixed; left: 16px; top: -100px; z-index: 100; background: var(--blue);
color: white; padding: 12px 18px; border-radius: 10px; }
.skip-link:focus { top: 12px; }
header { min-height: 76px; background: white; border-bottom: 1px solid var(--line); padding:
12px max(24px, calc((100vw - 1340px) / 2)); display: flex; align-items: center; gap: 25px; }
.brand { font-size: 26px; font-weight: 750; letter-spacing: -1px; display: flex; align-items:
center; white-space: nowrap; }
.brand > span:last-child { color: var(--blue); }
.brand-mark { background: var(--blue); color: white; width: 38px; height: 38px; display: grid;
 place-items: center; border-radius: 12px; margin-right: 11px; box-shadow: 0 3px 8px
#234bc51a; }
.brand-mark svg { width: 24px; height: 24px; }
.header-note { padding-left: 24px; border-left: 1px solid var(--line); color: var(--muted);
font-size: 14px; }
.header-actions { margin-left: auto; display: flex; align-items: center; gap: 20px; }
.prototype { font-size: 12px; color: #52647d; background: #f1f4f9; border: 1px solid #e1e7f0;
padding: 4px 10px; border-radius: 20px; }
.quiet { background: none; border: 1px solid transparent; padding: 10px 12px; color:
var(--blue); font-size: 14px; font-weight: 600; min-height: 44px; }
.quiet:hover:not(:disabled) { background: var(--blue-soft); }
.settings-button { display: flex; align-items: center; gap: 8px; color: var(--ink); }
.settings-button svg { width: 20px; height: 20px; }
main { max-width: 1388px; margin: auto; padding: 30px 24px 0; }
.intro { display: flex; align-items: center; justify-content: space-between; gap: 24px;
margin-bottom: 26px; }
.eyebrow { font-size: 12px; font-weight: 750; color: var(--blue); letter-spacing: 1.5px;
margin: 0 0 8px; }
h1 { font-size: clamp(28px, 3vw, 38px); font-weight: 750; line-height: 1.15; letter-spacing:
-1.5px; margin: 0 0 10px; }
.intro p:not(.eyebrow) { margin: 0; color: var(--muted); }
h2 { font-size: 23px; line-height: 1.3; letter-spacing: -.65px; margin: 0; }
h3 { font-size: 18px; line-height: 1.4; margin: 0; }
.intro-tools { display: flex; align-items: end; gap: 12px; }
.language { width: 228px; display: grid; gap: 7px; }
.language label { font-size: 14px; font-weight: 600; }
select { min-height: 46px; border: 1px solid #c5d1e2; background: white; border-radius: 11px;
padding: 10px 12px; width: 100%; color: var(--ink); }
.camera-jump.secondary, .board-jump.quiet { display: none; }
.layout { display: grid; grid-template-columns: 290px minmax(0, 1fr); gap: 24px; align-items:
start; }
.control-panel { background: white; border: 1px solid var(--line); border-radius:
var(--radius); padding: 22px; scroll-margin-top: 20px; }
.control-title { display: flex; justify-content: space-between; align-items: center; gap: 8px;
 }
.control-title h2 { font-size: 18px; letter-spacing: -.4px; }
.local-tag { font-size: 12px; color: #36598b; background: #eef4fc; border: 1px solid #dce6f5;
padding: 4px 7px; border-radius: 6px; white-space: nowrap; }
.camera-preview { aspect-ratio: 4/3; background: var(--navy); border-radius: 14px; overflow:
hidden; margin: 20px 0 14px; position: relative; }
.camera-preview video { width: 100%; height: 100%; object-fit: cover; transform: scaleX(-1);
display: block; }
#camera-placeholder { position: absolute; inset: 0; display: flex; flex-direction: column;
align-items: center; justify-content: center; text-align: center; color: #f0f5ff; }
#camera-placeholder > span { color: #8fa8d0; margin-bottom: 10px; }
#camera-placeholder svg { width: 44px; height: 44px; }
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dist/styles.css (continued)

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53    #camera-placeholder strong { font-size: 15px; font-weight: 600; }
54    #camera-placeholder p { font-size: 14px; color: #bccbe1; margin: 4px 0 20px; }
55    .camera-state { position: absolute; bottom: 0; left: 0; right: 0; background: #0e1e34dd;
+     padding: 6px 10px; font-size: 12px; color: #d1def2; text-align: center; }
56    .tracking-status { font-size: 14px; color: var(--muted); margin: 0 0 15px; min-height: 42px; }
57    .primary { background: var(--blue); border: 1px solid var(--blue); color: white; padding: 12px
+      16px; font-size: 15px; font-weight: 650; min-height: 48px; }
58    .primary:hover:not(:disabled), .confirm-button:hover:not(:disabled) { background: var(--blue-
+     dark); border-color: var(--blue-dark); }
59    .control-panel > .primary { width: 100%; }
60    .secondary { display: inline-flex; align-items: center; justify-content: center; min-height:
+     46px; background: white; border: 1px solid #c8d5e8; color: #284778; padding: 10px 12px; font-
+     size: 14px; font-weight: 600; border-radius: 11px; text-decoration: none; }
61    .secondary:hover:not(:disabled) { background: #f1f6fd; border-color: #a6bad7; }
62    .full { width: 100%; }
63    .camera-actions { display: grid; gap: 5px; margin-top: 9px; }
64    .compact-label { display: none; }
65    .camera-actions .quiet { color: var(--muted); }
66    .control-divider { height: 1px; background: var(--line); margin: 17px 0; }
67    .hint { font-size: 14px; color: var(--muted); margin: 10px 0 15px; }
68    .control-panel > .hint { font-size: 13px; margin-bottom: 16px; }
69    .instructions { font-size: 14px; color: var(--muted); margin: 16px 0; }
70    .instructions summary { font-size: 14px; font-weight: 600; color: #284778; cursor: pointer;
+     min-height: 36px; }
71    .instructions ol { padding-left: 21px; }
72    .instructions li { margin: 9px 0; }
73    .privacy { display: flex; align-items: start; gap: 9px; border-top: 1px solid var(--line);
+     padding-top: 16px; }
74    .privacy svg { color: #2c7182; width: 22px; height: 22px; flex-shrink: 0; }
75    .privacy strong { font-size: 14px; font-weight: 600; }
76    .privacy p { font-size: 13px; color: var(--muted); margin: 5px 0 0; }
77    .communication { background: white; border: 1px solid var(--line); border-radius:
+     var(--radius); padding: 26px; min-width: 0; box-shadow: 0 4px 20px #193a6605; }
78    .board-heading { display: flex; justify-content: space-between; align-items: center; gap:
+     16px; margin-bottom: 18px; }
79    .pause { background: #f4f7fc; border: 1px solid #d5dfed; color: #314b71; font-size: 14px;
+     font-weight: 600; padding: 10px 13px; min-height: 44px; max-width: 170px; }
80    .pause:hover:not(:disabled) { background: #e8effa; }
81    .board-state { display: flex; justify-content: space-between; gap: 12px; margin: 0 0 17px;
+     font-size: 14px; color: var(--muted); }
82    #hold-label { white-space: nowrap; }
83    .tracking-surface { position: relative; }
84    .message-grid { display: grid; grid-template-columns: repeat(3, minmax(0, 1fr)); gap: 14px; }
85    .message-card { position: relative; min-height: 156px; background: #f7f9fd; border: 2px solid
+     #e1e7f1; border-radius: 16px; color: var(--ink); padding: 19px 16px; display: flex; flex-
+     direction: column; align-items: start; justify-content: center; text-align: start; gap: 13px;
+     overflow: hidden; }
86    .message-card:hover:not(:disabled) { background: #eef3ff; border-color: #9cb6ee; }
87    .message-icon { width: 40px; height: 40px; border-radius: 12px; background: #e7eefe; color:
+     var(--blue); display: grid; place-items: center; flex-shrink: 0; }
88    .message-icon svg { width: 23px; height: 23px; stroke-width: 2.2; }
89    .message-card:first-child .message-icon { background: #e0f2eb; color: #166c50; }
90    .message-card:nth-child(2) .message-icon { background: #fce8e9; color: #ac3544; }
91    .message-card:nth-child(5) .message-icon { background: #e8edf7; color: #526685; }
92    .message-card:nth-child(6) .message-icon { background: #e0f1f5; color: #206479; }
93    .message-text { font-size: 18px; font-weight: 650; line-height: 1.4; overflow-wrap: anywhere;
+     }
94    .card-hint { font-size: 12px; letter-spacing: .4px; color: #52647d; font-weight: 600; }
95    .message-grid[lang=en] .card-hint { display: none; }
96    .message-card.hover-target, .confirmation button.hover-target, .pause.hover-target, .output-
+     actions button.hover-target { border-color: #b8860b; box-shadow: inset 0 0 0 2px #ffda79;
+     background: #fff4d7; color: #172641; }
97    .confirmation { padding: 30px 24px; border: 2px solid #adc2ee; border-radius: 16px;
+     background: #f0f5ff; text-align: center; }
98    .confirmation .eyebrow { color: #365998; }
99    .confirmation h3 { font-size: 18px; color: #526785; font-weight: 500; }
100   #selected-message { font-size: clamp(24px, 2.6vw, 32px); line-height: 1.5; font-weight: 700;
+     color: #173979; overflow-wrap: anywhere; margin: 14px 0; }
101   .voice-note { font-size: 14px; color: var(--muted); margin: 0 0 22px; }
102   .confirm-actions { display: flex; justify-content: center; gap: 14px; }
103   .confirm-actions button { min-height: 54px; padding: 13px 24px; font-size: 17px; font-weight:
+     650; border-radius: 12px; }
104   .confirm-button { background: var(--blue); border: 2px solid var(--blue); color: white; }
105   .cancel-button { background: white; border: 2px solid #bdcde5; color: #314b71; }
106   .cancel-button:hover { background: #e9f0fd; }
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dist/styles.css (continued)

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107   .message-output { margin-top: 18px; padding: 17px 20px; background: #eef3fb; border: 1px solid
+      #dce5f2; border-radius: 14px; }
108   .output-label { display: flex; align-items: center; justify-content: space-between; gap: 12px;
+      }
109   .output-label h3 { font-size: 12px; font-weight: 700; color: #4d6383; text-transform:
+     uppercase; letter-spacing: 1px; }
110   .output-actions { display: flex; gap: 2px; flex-shrink: 0; }
111   .output-actions button { padding: 8px; font-size: 14px; }
112   #output { font-size: clamp(24px, 2.2vw, 32px); line-height: 1.5; font-weight: 700; margin: 8px
+      0; color: #173979; overflow-wrap: anywhere; }
113   #output.empty { font-size: 20px; font-weight: 400; color: #536782; }
114   #output-status { font-size: 14px; color: var(--muted); }
115   .below-board { display: flex; justify-content: space-between; align-items: center; gap: 12px;
+     margin-top: 12px; font-size: 14px; color: var(--muted); }
116   .below-board .quiet { padding-inline: 8px; }
117   .head-pointer { position: fixed; width: 44px; height: 44px; transform: translate(-50%, -50%);
+     pointer-events: none; z-index: 80; filter: drop-shadow(0 1px 3px #0008); }
118   .head-pointer svg { width: 44px; height: 44px; transform: rotate(-90deg); }
119   .head-pointer circle { fill: none; stroke: var(--yellow); stroke-width: 4; stroke-dasharray:
+     113.1; stroke-dashoffset: 113.1; stroke-linecap: round; }
120   .head-pointer .ring-bg { stroke: #172b4999; stroke-dashoffset: 0; }
121   .head-pointer > span { position: absolute; left: 17px; top: 17px; width: 10px; height: 10px;
+     background: var(--yellow); border-radius: 50%; border: 2px solid white; }
122   .draft-open .message-grid { display: none; }
123   .draft-open .confirmation { min-height: 270px; }
124   .paused #pointer { opacity: .3; }
125   footer { display: flex; justify-content: space-between; gap: 20px; color: var(--muted); font-
+     size: 12px; padding: 22px 0 26px; }
126   dialog { max-width: 490px; width: calc(100% - 32px); max-height: calc(100dvh - 32px);
+     overflow-y: auto; overscroll-behavior: contain; border: 1px solid #cbd8eb; border-radius:
+     20px; padding: 28px; color: var(--ink); box-shadow: 0 24px 90px #0e24454d; }
127   dialog::backdrop { background: #14254080; backdrop-filter: blur(3px); }
128   dialog p { font-size: 16px; color: var(--muted); margin: 13px 0 21px; }
129   .dialog-heading { display: flex; gap: 12px; justify-content: space-between; align-items:
+     center; margin-bottom: 10px; }
130   .close { width: 44px; height: 44px; flex-shrink: 0; background: #f2f5fa; border: 0; color:
+     #52647d; font-size: 28px; }
131   .slider-label { display: flex; justify-content: space-between; gap: 15px; font-size: 15px;
+     font-weight: 600; margin: 19px 0 8px; }
132   .slider-label output { font-size: 14px; font-weight: 500; color: var(--muted); white-space:
+     nowrap; }
133   input[type=range] { width: 100%; height: 32px; accent-color: var(--blue); cursor: pointer; }
134   .check { display: flex; gap: 10px; align-items: center; font-size: 15px; margin-top: 20px; }
135   .check input { accent-color: var(--blue); width: 20px; height: 20px; flex-shrink: 0; }
136   dialog .hint { font-size: 14px; }
137   #settings-done, #save-phrases { width: 100%; margin-top: 6px; }
138   .calibration-diagram { height: 160px; border: 1px solid var(--line); border-radius: 14px;
+     background: #eff4fc; position: relative; margin: 20px 0; }
139   .cal-dot { width: 27px; height: 27px; background: var(--blue); border: 5px solid #bacdf8;
+     border-radius: 50%; position: absolute; left: 50%; top: 50%; transform: translate(-50%, -50%);
+      transition: left .3s, top .3s; z-index: 1; }
140   .cal-center { position: absolute; left: 50%; top: 50%; transform: translate(-50%, -50%);
+     color: #bdcbe1; font-size: 36px; }
141   .cal-progress { height: 7px; background: #e6edf8; border-radius: 5px; overflow: hidden; }
142   .cal-progress span { display: block; background: var(--blue); width: 0%; height: 100%; }
143   #calibration-dialog button { width: 100%; }
144   #phrase-fields { display: grid; gap: 14px; }
145   #phrase-fields label { display: grid; gap: 6px; font-size: 12px; font-weight: 600; color:
+     var(--muted); }
146   #phrase-fields input { width: 100%; min-height: 48px; border: 1px solid #c5d1e2; padding: 11px
+      12px; border-radius: 10px; font-size: 16px; }
147   [lang=ur], [lang=ar] { line-height: 1.8; }
148   /* Keep the board in view while a pointer is being used. */
149   .control-active main { padding-top: 18px; }
150   .control-active .intro { margin-bottom: 18px; }
151   .control-active .intro .eyebrow { display: none; }
152   .control-active h1 { font-size: 28px; margin-bottom: 4px; }
153   .control-active .intro p { font-size: 14px; }
154   .control-active .message-card { min-height: 120px; padding: 14px; gap: 10px; }
155   .control-active .message-icon { width: 34px; height: 34px; border-radius: 10px; }
156   .control-active .message-text { font-size: 17px; }
157   .control-active .communication { padding: 22px; }
158   .control-active .message-output { margin-top: 14px; padding: 12px 16px; }
159   .control-active #output { font-size: 24px; margin: 4px 0; }
160   .control-active #output.empty { font-size: 18px; }
```

dist/styles.css (continued)

```
@media (max-width: 1100px) {
  .header-note { display: none; }
  .layout { grid-template-columns: 260px minmax(0, 1fr); gap: 18px; }
  .control-panel { padding: 18px; }
  .communication { padding: 22px; }
  .board-state { flex-wrap: wrap; gap: 4px 12px; }
  .message-card { padding: 16px 12px; }
  .message-text { font-size: 17px; }
  .local-tag { font-size: 11px; }
}
@media (max-width: 900px) {
  main { padding: 24px 20px 0; }
  .layout { grid-template-columns: 1fr; gap: 18px; }
  .communication { order: -1; }
  .control-panel { display: grid; grid-template-columns: 200px minmax(0, 1fr); gap: 12px 20px;
+  }
  .control-title { grid-column: 1 / -1; }
  .camera-preview { grid-row: 2 / 5; margin: 0; }
  .tracking-status { margin: 0; min-height: 0; align-self: center; }
  .control-panel > .primary, .camera-actions { grid-column: 2; }
  .camera-actions { grid-template-columns: 1fr 1fr; margin: 0; }
  .control-divider, #practice, .control-panel > .hint, .instructions, .privacy, .board-jump {
+ grid-column: 1 / -1; }
  .control-divider { margin: 5px 0; }
  .control-panel > .hint, .instructions { margin: 0; }
  .privacy { padding-top: 14px; }
  .camera-jump.secondary { display: inline-flex; }
  .board-jump.quiet { display: block; text-decoration: none; text-align: center; }
  .control-active .communication { order: 0; }
  .control-active .control-panel { display: grid; grid-template-columns: 58px minmax(0, 1fr)
+ auto; gap: 6px 12px; padding: 12px 14px; align-items: center; }
  .control-active .control-panel > * { display: none; }
  .control-active .control-panel .tracking-status { display: block; grid-column: 2 / 4; font-
+ size: 14px; }
  .control-active .camera-preview { display: block; grid-row: 1 / 3; grid-column: 1; width:
+ 58px; height: 58px; aspect-ratio: auto; border-radius: 10px; }
  .control-active .camera-state, .control-active #camera-placeholder { display: none; }
  .control-active .control-panel > #start-camera { display: block; grid-column: 2; grid-row:
+ 2; width: auto; justify-self: start; min-height: 40px; padding: 7px 12px; font-size: 14px; }
  .control-active .camera-actions { display: flex; grid-column: 3; grid-row: 2; gap: 8px; }
  .control-active .camera-actions button { font-size: 14px; padding: 7px 10px; min-height:
+ 40px; }
  .control-active .camera-actions { grid-column: 2 / 4; justify-self: end; }
  .control-active #start-camera:disabled { display: none; }
  .control-active .camera-actions button:disabled { display: none; }
  .control-active .wide-label { display: none; }
  .control-active .compact-label { display: inline; }
  .control-active .control-panel:has(#start-camera:not(:disabled)) { grid-template-columns:
+ minmax(0, 1fr) auto; }
  .control-active .control-panel:has(#start-camera:not(:disabled)) .tracking-status { grid-
+ column: 1 / -1; }
  .control-active .control-panel:has(#start-camera:not(:disabled)) .camera-preview { display:
+ none; }
  .control-active .control-panel > #start-camera:not(:disabled) { grid-column: 1; }
  .control-active #start-camera:not(:disabled) ~ .camera-actions { display: none; }
  .control-active .camera-jump.secondary { display: none; }
  .control-active .intro > div:first-child { display: none; }
  .control-active .intro { justify-content: end; margin-bottom: 12px; }
  .control-active .language { display: flex; align-items: center; gap: 12px; width: auto; }
  .control-active .language select { width: 205px; }
  .control-active .layout { gap: 12px; }
}
@media (max-width: 600px) {
  header { min-height: 66px; padding: 10px 16px; gap: 12px; }
  .brand { font-size: 23px; }
  .brand-mark { width: 33px; height: 33px; margin-right: 8px; border-radius: 10px; }
  .brand-mark svg { width: 21px; height: 21px; }
  .prototype { display: none; }
  .header-actions { gap: 0; }
  .settings-button { padding: 8px 4px 8px 8px; }
  main { padding: 22px 14px 0; }
  .intro { display: block; margin-bottom: 18px; }
  h1 { font-size: 30px; }
  .intro p:not(.eyebrow) { font-size: 16px; }
  .intro-tools { margin-top: 18px; width: 100%; gap: 10px; }
```

dist/styles.css (continued)

```
226     .language { width: auto; flex: 1; min-width: 0; gap: 5px; }
227     .camera-jump { flex-shrink: 0; }
228     .communication { padding: 18px 14px; border-radius: 18px; }
229     .board-heading { align-items: start; gap: 10px; margin-bottom: 14px; }
230     .board-heading h2 { font-size: 21px; }
231     .board-heading .eyebrow { font-size: 11px; letter-spacing: 1.1px; }
232     .pause { max-width: 112px; padding: 9px; font-size: 14px; }
233     .pause .wide-label { display: none; }
234     .pause .compact-label { display: inline; }
235     .board-state { display: flex; flex-direction: column; gap: 4px; margin-bottom: 14px; font-
+     size: 14px; }
236     .message-grid { grid-template-columns: repeat(2, minmax(0, 1fr)); gap: 10px; }
237     .message-card { min-height: 148px; padding: 14px 12px; gap: 10px; border-radius: 13px; }
238     .message-text { font-size: 17px; }
239     .message-icon { width: 35px; height: 35px; border-radius: 10px; }
240     .message-icon svg { width: 21px; height: 21px; }
241     .message-output { padding: 14px; }
242     .output-label { flex-wrap: wrap; gap: 0 8px; }
243     .output-label h3 { font-size: 12px; letter-spacing: .6px; }
244     .output-actions { margin-left: auto; }
245     #output { font-size: 25px; }
246     #output.empty { font-size: 18px; }
247     .below-board { flex-wrap: wrap; gap: 0 8px; }
248     .below-board .quiet { margin-left: auto; }
249     .control-panel { grid-template-columns: 110px minmax(0, 1fr); gap: 12px; padding: 18px;
+     border-radius: 18px; }
250     .camera-preview { grid-row: 2 / 4; height: 130px; aspect-ratio: auto; }
251     #camera-placeholder svg { width: 32px; height: 32px; }
252     #camera-placeholder > span { margin-bottom: 5px; }
253     #camera-placeholder strong { font-size: 14px; padding: 0 8px; }
254     #camera-placeholder p { display: none; }
255     .camera-state { padding: 6px; font-size: 11px; }
256     .camera-actions { grid-column: 1 / -1; grid-template-columns: 1fr 1fr; }
257     .tracking-status { font-size: 14px; }
258     .control-panel > .primary { padding: 10px; font-size: 14px; }
259     .confirmation { padding: 24px 14px; }
260     #selected-message { font-size: 26px; }
261     .confirm-actions { gap: 10px; }
262     .confirm-actions button { font-size: 16px; padding: 12px 15px; }
263     dialog { padding: 22px 20px; border-radius: 18px; }
264     dialog h2 { font-size: 22px; }
265     footer { flex-direction: column; gap: 6px; padding: 20px 4px; }
266     .control-active header { min-height: 58px; padding-block: 6px; }
267     .control-active main { padding-top: 10px; }
268     .control-active .intro { display: flex; margin-bottom: 10px; }
269     .control-active .intro-tools { margin-top: 0; }
270     .control-active .language { justify-content: space-between; width: 100%; gap: 8px; }
271     .control-active .language label { font-size: 14px; }
272     .control-active .language select { width: 180px; min-height: 42px; padding-block: 7px; }
273     .control-active .control-panel { grid-template-columns: 46px minmax(0, 1fr) auto; padding:
+     9px 11px; gap: 4px 9px; }
274     .control-active .camera-preview { width: 46px; height: 58px; }
275     .control-active .tracking-status { font-size: 13px; }
276     .control-active .camera-actions button, .control-active .control-panel > #start-camera {
+     font-size: 13px; padding: 6px 9px; min-height: 38px; }
277     .control-active .communication { padding: 14px; }
278     .control-active .board-heading { margin-bottom: 8px; }
279     .control-active .board-heading h2 { font-size: 19px; }
280     .control-active .board-heading .eyebrow { display: none; }
281     .control-active .board-state { gap: 2px; margin-bottom: 10px; font-size: 13px; }
282     .control-active .message-card { min-height: 92px; padding: 11px; gap: 7px; }
283     .control-active .message-icon { width: 26px; height: 26px; border-radius: 7px; }
284     .control-active .message-icon svg { width: 17px; height: 17px; }
285     .control-active .message-text { font-size: 16px; }
286     .control-active .card-hint { display: none; }
287     .control-active .message-output { padding: 10px 12px; margin-top: 12px; }
288     .control-active .output-label h3 { font-size: 11px; letter-spacing: .5px; }
289     .control-active .output-actions button { min-height: 38px; padding: 6px; font-size: 13px; }
290     .control-active #output { font-size: 22px; margin: 2px 0; }
291     .control-active #output.empty { font-size: 16px; }
292     .control-active #output-status { font-size: 13px; }
293     .control-active .below-board { font-size: 13px; margin-top: 6px; }
294     .control-active .below-board .quiet { font-size: 13px; }
295     .control-active .confirmation { padding: 20px 14px; }
296     .control-active .confirmation .eyebrow { display: block; }
```

dist/styles.css (continued)

```
297   }
298   @media (max-width: 600px) and (max-height: 720px) {
299     .control-active main { padding-top: 6px; }
300     .control-active .intro { margin-bottom: 6px; }
301     .control-active .language label { position: absolute; width: 1px; height: 1px; overflow:
+     hidden; clip-path: inset(50%); }
302     .control-active .language { justify-content: end; }
303     .control-active .language select { min-height: 38px; }
304     .control-active .control-panel { display: flex; flex-wrap: wrap; gap: 4px 8px; padding: 8px
+     10px; }
305     .control-active .control-panel .tracking-status { font-size: 12px; flex: 1 1 100%; }
306     .control-active .camera-preview { display: none; }
307     .control-active .camera-actions { margin-left: auto; }
308     .control-active .layout { gap: 8px; }
309     .control-active .communication { padding: 10px 12px; }
310     .control-active .board-heading h2 { font-size: 17px; }
311     .control-active .board-heading { margin-bottom: 5px; }
312     .control-active .pause { min-height: 40px; padding: 8px; }
313     .control-active .message-icon { display: none; }
314     .control-active .message-grid { gap: 8px; }
315     .control-active .message-card { min-height: 65px; padding: 8px; }
316     .control-active .message-text { font-size: 14px; line-height: 1.35; }
317     .control-active .board-state { flex-direction: row; flex-wrap: wrap; gap: 0 8px; font-size:
+     12px; margin-bottom: 6px; }
318     .control-active .message-output { padding: 7px 9px; margin-top: 8px; }
319     .control-active .output-label { flex-wrap: nowrap; gap: 4px; }
320     .control-active .output-label h3 { font-size: 12px; letter-spacing: .2px; }
321     .control-active .output-actions button { font-size: 12px; min-height: 36px; padding: 5px; }
322     .control-active #output { font-size: 19px; }
323     .control-active #output.empty { font-size: 14px; }
324     .control-active #output-status { font-size: 12px; }
325     .control-active .below-board { margin-top: 0; }
326   }
327   @media (max-height: 540px) and (min-width: 601px) and (max-width: 900px) {
328     .control-active header { min-height: 50px; padding-block: 3px; }
329     .control-active .intro { display: none; }
330     .control-active .control-panel { display: flex; gap: 12px; padding: 8px 12px; }
331     .control-active .camera-preview { display: none; }
332     .control-active .tracking-status { margin-right: auto; }
333     .control-active .communication { padding: 12px 16px; }
334     .control-active .board-heading { margin-bottom: 6px; }
335     .control-active .board-heading h2 { font-size: 19px; }
336     .control-active .board-heading .eyebrow { display: none; }
337     .control-active .board-state { margin-bottom: 8px; font-size: 13px; }
338     .control-active .message-card { min-height: 67px; padding: 9px 12px; }
339     .control-active .message-icon, .control-active .card-hint { display: none; }
340     .control-active .message-text { font-size: 16px; }
341     .control-active .message-output { margin-top: 10px; }
342     .control-active .draft-open .confirmation { min-height: 0; padding: 10px 14px; }
343     .control-active .confirmation .eyebrow { display: none; }
344     .control-active .confirmation h3 { font-size: 14px; }
345     .control-active #selected-message { font-size: 22px; margin: 4px 0; }
346     .control-active .voice-note { font-size: 13px; margin: 0 0 8px; }
347     .control-active .confirm-actions button { min-height: 44px; padding: 8px 16px; font-size:
+     16px; }
348   }
349   @media (prefers-reduced-motion: reduce) { * { transition: none !important; scroll-behavior:
+     auto !important; } }
350   @media (forced-colors: active) { .message-card, .primary, .secondary, .pause, .confirm-button,
+      .cancel-button { border: 2px solid ButtonText; } }
```

dist/app.mjs

```
import {clamp,makeCalibration,mapPointer,DwellGate} from './core.mjs';
import {waitForCameraFrames} from './camera.mjs';
import {SteadyPointer} from './pointer.mjs';
import {SpeechPlayer,speechLanguages} from './speech.mjs';
const $=id=>document.getElementById(id);
const ids=['yes','no','again','write','break','caregiver'];
const packs={
 en:['Yes.','No.','Please explain that again.','Please write it down.','I need a break.','Please call my caregiver.'],
 ur:['\u062c\u06cc \u06c1\u0627\u06ba\u06d4','\u0646\u06c1\u06cc\u06ba\u06d4','\u0628\u0631\u0627\u06c1 \u06a9\u0631\u0645 \u062f\u0648\u0628\u0627\u0631\u06c1 \u0633\u0645\u062c\u06be\u0627\u0626\u06cc\u06ba\u06d4','\u0628\u0631\u0627\u06c1 \u06a9\u0631\u0645 \u0644\u06a9\u06be \u062f\u06cc\u06ba\u06d4','\u0645\u062c\u06be\u06d2 \u0648\u0642\u0641\u06c1 \u0686\u0627\u06c1\u06cc\u06d2\u06d4','\u0628\u0631\u0627\u06c1 \u06a9\u0631\u0645 \u0645\u06cc\u0631\u06d2 \u062f\u06cc\u06a9\u06be \u0628\u06be\u0627\u0644 \u06a9\u0631\u0646\u06d2 \u0648\u0627\u0644\u06d2 \u06a9\u0648 \u0628\u0644\u0627\u06a9\u06d4'],
 ar:['\u0646\u064e\u0639\u064e\u0645\u0652.','\u0644\u064e\u0627.','\u064a\u064f\u0631\u0652\u062c\u064e\u0649 \u0627\u0644\u0634\u0651\u064e\u0631\u0652\u062d\u064f \u0645\u064e\u0631\u0651\u064e\u0629\u064b \u0623\u064f\u062e\u0652\u0631\u064e\u0649.','\u064a\u064f\u0631\u0652\u062c\u064e\u0649 \u0643\u0650\u062a\u064e\u0627\u0628\u064e\u0629\u064f \u0630\u064e\u0644\u0650\u0643\u064e.','\u0623\u064e\u062d\u0652\u062a\u064e\u0627\u062c\u064f \u0625\u0650\u0644\u064e\u0649 \u0627\u0650\u0633\u0652\u062a\u0650\u0631\u064e\u0627\u062d\u064e\u0629\u064d.','\u064a\u064f\u0631\u0652\u062c\u064e\u0649 \u0627\u0650\u0633\u0652\u062a\u0650\u0639\u064e\u0627\u0621\u064f \u0645\u064e\u0646\u0652 \u064a\u064e\u0639\u0652\u062a\u064e\u0646\u0650\u064a \u0628\u0650\u064a.'],
 hi:['\u0939\u093e\u0901\u0964','\u0928\u0939\u0940\u0902\u0964','\u0915\u0943\u092a\u092f\u093e \u092b\u093f\u0930 \u0938\u0947 \u0938\u092e\u091d\u093e\u090f\u0901\u0964','\u0915\u0943\u092a\u092f\u093e \u0907\u0938\u0947 \u0932\u093f\u0916 \u0926\u0947\u0902\u0964','\u092e\u0941\u091d\u0947 \u0925\u094b\u0921\u093c\u093e \u0938\u092e\u092f \u0905\u0915\u0947\u0932\u093e \u091b\u094b\u0921\u093c\u0947\u0902\u0964','\u0915\u0943\u092a\u092f\u093e \u092e\u0947\u0930\u0947 \u0926\u094b\u0938\u094d\u0924 \u0915\u094b \u092c\u0941\u0932\u093e\u090f\u0901\u0964'],
 es:['S\xed.','No.','Por favor, expl\xedquelo otra vez.','Por favor, escr\xedbalo.','Necesito un descanso.','Por favor, llame a mi cuidador.'],
 fr:['Oui.','Non.','Veuillez expliquer \xe0 nouveau.','Veuillez l\u2019\xe9crire.','J\u2019ai besoin d\u2019une pause.','Veuillez appeler mon aidant.'],
 de:['Ja.','Nein.','Bitte erkl\xe4ren Sie das noch einmal.','Bitte schreiben Sie es auf.','Ich brauche eine Pause.','Bitte rufen Sie meine Betreuungsperson.'],
 zh:['\u662f\u7684\u3002','\u4e0d\u662f\u3002','\u8bf7\u518d\u89e3\u91ca\u4e00\u904d\u3002','\u8bf7\u5199\u4e0b\u6765\u3002','\u6211\u9700\u8981\u4f11\u606f\u4e00\u4e0b\u3002','\u8bf7\u53eb\u6211\u7684\u7167\u62a4\u8005\u3002']
};
const originalPacks=structuredClone(packs);
const symbols=['<path d="m5 12 4 4L19 6"/>','<path d="m6 6 12 12M18 6 6 18"/>','<path d="M4 10a8 8 0 1 1 2 8M4 4v6h6"/>','<path d="m15 4 5 5L9 20H4v-5L15 4ZM13 6l5 5"/>','<path d="M8 5v14M16 5v14"/>','<circle cx="12" cy="7" r="3"/><path d="M5 21v-3a7 7 0 0 1 14 0v3M8 21v-4M16 21v-4"/>'];
const hints=['YES','NO','EXPLAIN AGAIN','WRITE IT DOWN','TAKE A BREAK','CALL MY CAREGIVER'];
let lang='en',draft=null,confirmed=null,mode='click',paused=false,stream=null,worker=null,workerReady=false,inflight=false;
let cameraEpoch=0,lastTracking=0,frameStarted=0,faceCount=0,nose=null,calibration=null,calState=null,point=null,mousePoint=null;
let sensitivity=.8,alpha=.12,reverseX=false,releaseRequired=false,cooldown=0,frameTimer=null;
const steadyPointer=new SteadyPointer();
const speech=new SpeechPlayer({onStatus:message=>{$('output-status').textContent=message;}});
const gate=new DwellGate(1500),pointer=$('pointer'),ring=$('pointer-ring');
const dialogs=[$('settings-dialog'),$('calibration-dialog'),$('phrases-dialog')];
const dialogOpen=()=>dialogs.some(d=>d.open);
function status(message){if($('tracking-status').textContent!==message)$('tracking-status').textContent=message;}
function resetDwell(requireRelease=false){gate.reset();ring.style.strokeDashoffset='113.1';document.querySelectorAll('.hover-target').forEach(el=>el.classList.remove('hover-target'));releaseRequired=requireRelease;}
function render(){
 const grid=$('message-grid');grid.replaceChildren();grid.dir=['ur','ar'].includes(lang)?'rtl':'ltr';grid.lang=lang;
 ids.forEach((id,i)=>{const b=document.createElement('button');b.className='message-card';b.dataset.target=id;b.dataset.messageId=id;b.disabled=!!draft;
  const icon=document.createElement('span');icon.className='message-icon';icon.innerHTML='<svg viewBox="0 0 24 24">'+symbols[i]+'</svg>';icon.setAttribute('aria-hidden','true');
```

dist/app.mjs (continued)

```
34      const text=document.createElement('strong');text.className='message-
+     text';text.textContent=packs[lang][i];const
+     hint=document.createElement('span');hint.className='card-
+     hint';hint.textContent=packs[lang][i]===originalPacks[lang][i]?hints[i]:'YOUR MESSAGE';
35      b.append(icon,text,hint);b.addEventListener('click',()=>activate(id));grid.append(b);
36     });
37    }
38    function voiceInfo(){const device=speech.info(lang)==='device';$('voice-
+     settings').textContent=device?'A local device voice is available. It will be used
+     first.':'Free built-in '+speechLanguages[lang]+' voice. It works on this device and sounds
+     synthetic.';if(lang==='ar')$('voice-settings').textContent+=' Arabic pronunciation works best
+     with vowel marks.';if(draft)$('voice-note').textContent=device?'Confirm to display and read
+     this message aloud.':'Confirm to display and read with the free built-in voice. Pronunciation
+     may vary.';}
39    function stage(id){
40     if(!ids.includes(id))throw new Error('Unknown message ID.');
41     if(dialogOpen())throw new Error('Close the open dialog before selecting a message.');
42     if(paused)throw new Error('Resume control before selecting a message.');
43     if(draft)throw new Error('Confirm or cancel the current message first.');
44     draft={id,text:packs[lang][ids.indexOf(id)],lang};$('confirmation').hidden=false;$('selected-
+     message').textContent=draft.text;$('selected-message').lang=lang;$('selected-
+     message').dir=['ur','ar'].includes(lang)?'rtl':'ltr';
45     $('surface').classList.add('draft-open');$('selection-status').textContent='Message selected.
+      Confirm or cancel it.';render();voiceInfo();cooldown=performance.now()+600;resetDwell(true);
46     if(mode!=='camera')requestAnimationFrame(()=>{if(!draft)return;$('confirmation').scrollIntoVi
+     ew({block:'nearest'});if(mode==='click')$('confirm').focus({preventScroll:true});});
47    }
48    function dismiss(){const previous=draft?.id;draft=null;$('confirmation').hidden=true;$('surfac
+     e').classList.remove('draft-open');$('selection-status').textContent='No message selected.';re
+     nder();cooldown=performance.now()+600;resetDwell(true);if(previous&&mode==='click')document.qu
+     erySelector('[data-message-id="'+previous+'"]').focus({preventScroll:true});}
49    function speak(message){speech.speak(message);}
50    function confirm(){if(!draft)return;confirmed={...draft};$('output').classList.remove('empty')
+     ;$('output').textContent=confirmed.text;$('output').lang=confirmed.lang;$('output').dir=['ur',
+     'ar'].includes(confirmed.lang)?'rtl':'ltr';$('repeat').disabled=false;$('clear').disabled=fals
+     e;dismiss();speak(confirmed);}
51    function clear(){speech.cancel();confirmed=null;$('output').classList.add('empty');$('output')
+     .textContent='Your message will appear
+     here.';$('output').lang='en';$('output').dir='ltr';$('output-status').textContent='Nothing is
+     spoken until you
+     confirm.';$('repeat').disabled=true;$('clear').disabled=true;resetDwell(true);}
52    function pauseLabel(){const button=$('pause');button.setAttribute('aria-label',paused?'Resume
+     head control':'Pause head control');button.querySelector('.wide-
+     label').textContent=paused?'Resume head control':'Pause head
+     control';button.querySelector('.compact-label').textContent=paused?'Resume':'Pause';}
53    function togglePause(){paused=!paused;document.body.classList.toggle('paused',paused);pauseLab
+     el();$('mode-label').textContent=paused?'Selection paused. Move to Resume to
+     continue.':mode==='camera'?'Head control active.':'Practice mode \xb7 tap, click, or hold with
+      a mouse.';resetDwell(true);}
54    function activate(target){try{if(ids.includes(target))stage(target);else
+     if(target==='confirm')confirm();else if(target==='cancel')dismiss();else
+     if(target==='pause')togglePause();else if(target==='repeat')speak(confirmed);else
+     if(target==='clear')clear();}catch(error){$('selection-status').textContent=error.message;}}
55    function pointerArea(){const
+     rect=document.querySelector('.communication').getBoundingClientRect();const top=Math.max(0,rec
+     t.top),bottom=Math.min(innerHeight,rect.bottom);return{left:rect.left+24,top:top+24,width:Math
+     .max(1,rect.width-48),height:Math.max(1,bottom-top-48)};}
56    function setPoint(normalized){const area=pointerArea();point={x:area.left+normalized.x*area.wi
+     dth,y:area.top+normalized.y*area.height};pointer.style.left=point.x+'px';pointer.style.top=poi
+     nt.y+'px';pointer.hidden=false;}
57    function renderPointer(now){
58     const
+     valid=mode==='practice'?!!mousePoint:mode==='camera'&&!!calibration&&faceCount===1&&now-
+     lastTracking<350;
59     if(!valid||dialogOpen()||document.hidden){pointer.hidden=true;resetDwell();if(mode==='camera'
+     &&!valid)steadyPointer.reset();return;}
60     if(mode==='practice'){point=mousePoint;pointer.style.left=point.x+'px';pointer.style.top=poin
+     t.y+'px';pointer.hidden=false;}
61     else{const
+     smoothed=steadyPointer.step(now,{...pointerArea(),alpha});if(smoothed)setPoint(smoothed);}
62     if(!point){pointer.hidden=true;return;}
63     let targetElement=document.elementFromPoint(point.x,point.y)?.closest('[data-target]');
64     let target=targetElement&&!targetElement.disabled?targetElement.dataset.target:null;
65     if(paused&&target!=='pause')target=null;
66     if(draft&&ids.includes(target))target=null;
```

dist/app.mjs (continued)

```
67      if(releaseRequired){if(!target)releaseRequired=false;resetDwell(releaseRequired);return;}
68      if(now<cooldown){resetDwell();return;}
69      const result=gate.update(target,now,true);
70      document.querySelectorAll('.hover-target').forEach(el=>el.classList.remove('hover-
+     target'));if(target)targetElement.classList.add('hover-
+     target');ring.style.strokeDashoffset=String(113.1*(1-result.progress));
71      if(result.fire)activate(result.fire);
72    }
73    function animate(now){renderPointer(now);requestAnimationFrame(animate);}
74    function showDialog(id){resetDwell();$(id).showModal();}
75    function applyTracking(data){
76      inflight=false;const now=performance.now();lastTracking=data.timestamp;faceCount=now-
+     data.timestamp<350?data.count:0;nose=faceCount===1?data.nose:null;
77      if(faceCount!==1){point=null;steadyPointer.reset();resetDwell();const delayed=now-
+     data.timestamp>=350;const message=faceCount>1?'Keep only one face in the camera
+     view.':delayed?'Tracking is delayed. Close other busy apps or try Chrome / Edge.':'Face not
+     found. Face the camera in good light.';status(message+' Selection is stopped.');if(calState){c
+     alState.since=null;calState.samples[steps[calState.index].key]=[];$('calibration-
+     step').textContent=`Step ${calState.index+1} of 5 \xb7 ${message}`;$('cal-
+     progress').style.width='0%';}return;}
78      if(calState){collectCalibration(lastTracking);return;}
79      if(!calibration){status('Face found. Set your movement range to enable head
+     control.');return;}
80      status(paused?'Face found. Selection is paused.':'Face tracked \xb7 head control ready.');
81      const mapped=mapPointer(nose,calibration,sensitivity,reverseX);if(!mapped)return;
82      steadyPointer.push(mapped,pointerArea());
83    }
84    async function sendFrame(epoch){
85      if(epoch!==cameraEpoch||mode!=='camera'||!workerReady)return;
86      if(inflight&&performance.now()-frameStarted>8000){stopCamera('The face tracker stopped
+     responding. Try again, or open the demo in Chrome or Edge.');return;}
87      if(!inflight&&$('video').readyState>=2){
88        inflight=true;frameStarted=performance.now();try{const bitmap=await createImageBitmap($('vid
+     eo'));if(epoch!==cameraEpoch||!worker){bitmap.close();inflight=false;return;}worker.postMessag
+     e({kind:'frame',bitmap,timestamp:performance.now()},[bitmap]);}catch{inflight=false;status('Ca
+     mera frame unavailable. Selection is stopped.');faceCount=0;}
89      }
90      frameTimer=setTimeout(()=>sendFrame(epoch),85);
91    }
92    function stopCamera(message='Camera is off. Click a card, or try practice mode.'){
93      cameraEpoch++;steadyPointer.reset();clearTimeout(frameTimer);frameTimer=null;stream?.getTrack
+     s().forEach(track=>track.stop());stream=null;worker?.terminate();worker=null;workerReady=false
+     ;inflight=false;$('video').srcObject=null;document.body.classList.remove('control-active');
94      if(calState)cancelCalibration();faceCount=0;lastTracking=0;nose=null;calibration=null;point=n
+     ull;mode='click';paused=false;document.body.classList.remove('paused');pointer.hidden=true;$('
+     camera-placeholder').hidden=false;$('start-camera').disabled=false;$('start-
+     camera').textContent='Enable camera';$('calibrate').disabled=true;$('stop-
+     camera').disabled=true;$('pause').disabled=true;pauseLabel();$('camera-
+     state').textContent='Camera off';$('mode-label').textContent='Tap or click a message to
+     begin.';resetDwell();status(message);
95    }
96    async function startCamera(){
97      void speech.prepare();
98      stopCamera();const epoch=cameraEpoch;mode='camera';mousePoint=null;$('start-
+     camera').disabled=true;$('stop-camera').disabled=false;$('start-
+     camera').textContent='Starting\u2026';status('Loading the face model on your device\u2026');
99      try{
100       if(!window.isSecureContext)throw new Error('Camera needs HTTPS or localhost. Open the hosted
+      demo.');
101       if(!navigator.mediaDevices?.getUserMedia||!window.createImageBitmap)throw new Error('This
+     browser does not support the camera features. Try a recent Chrome or Edge browser.');
102       worker=new Worker('./vision-worker.js');
103       await new Promise((resolve,reject)=>{
104         const timeout=setTimeout(()=>reject(new Error('Model loading timed out. Check your
+     connection and try again.')),90000);
105         worker.onerror=()=>{clearTimeout(timeout);reject(new Error('The local face model could not
+     load. Practice mode is still available.'));};
106         worker.onmessage=({data})=>{if(epoch!==cameraEpoch)return;if(data.kind==='ready'){clearTime
+     out(timeout);workerReady=true;resolve();}else
+     if(data.kind==='error'){clearTimeout(timeout);reject(new Error(data.message));}};
107         worker.postMessage({kind:'init'});
108       });
109       if(epoch!==cameraEpoch)return;status('Allow camera access in your browser to continue.');
110       const acquired=await navigator.mediaDevices.getUserMedia({video:{width:{ideal:640},height:{i
+     deal:480},facingMode:'user'},audio:false});
```

dist/app.mjs (continued)

```
111     if(epoch!==cameraEpoch){acquired.getTracks().forEach(t=>t.stop());return;}stream=acquired;$(
+     'video').srcObject=stream;$('camera-placeholder').hidden=true;$('camera-
+     state').textContent='Camera connected \xb7 waiting for frames';status('Camera connected.
+     Waiting for video frames\u2026');
112     await waitForCameraFrames($('video'),{isCurrent:()=>epoch===cameraEpoch});if(epoch!==cameraE
+     poch)return;
113     $('camera-placeholder').hidden=true;$('start-camera').textContent='Camera enabled';$('stop-
+     camera').disabled=false;$('calibrate').disabled=false;$('camera-state').textContent='Local
+     camera \xb7 no upload';
114     worker.onmessage=({data})=>{if(epoch!==cameraEpoch)return;if(data.kind==='tracking')applyTra
+     cking(data);else if(data.kind==='error')stopCamera('Tracking stopped: '+data.message);};
115     worker.onerror=()=>stopCamera('Tracking stopped. Try again or use practice mode.');
116     stream.getVideoTracks()[0].addEventListener('ended',()=>{if(epoch===cameraEpoch)stopCamera('
+     Camera access ended. Selection is stopped.');});
117     beginCalibration();sendFrame(epoch);
118    }catch(error){if(epoch!==cameraEpoch)return;const
+     message=error.name==='NotAllowedError'?'Camera permission was declined. Practice mode still
+     works.':error.message;stopCamera(message);}
119   }
120   const steps=[{key:'center',title:'Look straight ahead.',text:'Keep your head centered and
+     relaxed.',x:50,y:50},{key:'left',title:'Move gently to your left.',text:'Move or turn your
+     head left only as far as comfortable. Hold there.',x:15,y:50},{key:'right',title:'Move gently
+     to your right.',text:'Move or turn your head right only as far as comfortable. Hold
+     there.',x:85,y:50},{key:'up',title:'Move gently upward.',text:'Raise or tilt your head
+     slightly, only as far as comfortable. Hold there.',x:50,y:15},{key:'down',title:'Move gently
+     downward.',text:'Lower or tilt your head slightly, only as far as comfortable. Hold
+     there.',x:50,y:85}];
121   function beginCalibration(){if(!stream)return;dismiss();calibration=null;point=null;steadyPoin
+     ter.reset();$('pause').disabled=true;calState={index:0,since:null,samples:{}};showDialog('cali
+     bration-dialog');showStep();}
122   function showStep(){const
+     step=steps[calState.index];calState.samples[step.key]=[];calState.since=null;$('calibration-
+     title').textContent=step.title;$('calibration-text').textContent=step.text;$('calibration-
+     step').textContent=`Step ${calState.index+1} of 5 \xb7 wait for one visible face`;$('cal-
+     dot').style.left=step.x+'%';$('cal-dot').style.top=step.y+'%';$('cal-
+     progress').style.width='0%';}
123   function collectCalibration(now){
124    if(!calState)return;if(calState.since===null)calState.since=now;
125    const elapsed=now-calState.since,step=steps[calState.index];$('calibration-
+     step').textContent=`Step ${calState.index+1} of 5 \xb7 ${elapsed<1000?'get
+     comfortable\u2026':'hold this position\u2026'}`;$('cal-
+     progress').style.width=clamp(elapsed/3000*100,0,100)+'%';
126    if(elapsed>=1000)calState.samples[step.key].push({...nose});
127    if(elapsed>=3000){
128     if(++calState.index<steps.length){showStep();return;}
129     try{calibration=makeCalibration(calState.samples);calState=null;$('calibration-
+     dialog').close();paused=false;document.body.classList.add('control-
+     active');window.scrollTo(0,0);$('pause').disabled=false;$('mode-label').textContent='Head
+     control active.';status('Movement range set. Move your head to guide the
+     pointer.');resetDwell(true);}
130     catch(error){calState=null;$('calibration-
+     dialog').close();status(error.message);calibration=null;}
131    }
132   }
133   function cancelCalibration(){calState=null;if($('calibration-dialog').open)$('calibration-
+     dialog').close();status('Setup canceled. Set your movement range when you\u2019re
+     ready.');resetDwell();}
134   $('start-camera').addEventListener('click',startCamera);$('stop-camera').addEventListener('cli
+     ck',()=>stopCamera());$('calibrate').addEventListener('click',beginCalibration);$('cancel-
+     calibration').addEventListener('click',cancelCalibration);$('calibration-
+     dialog').addEventListener('cancel',()=>{calState=null;status('Setup canceled. Set your
+     movement range when you\u2019re ready.');});
135   $('practice').addEventListener('click',()=>{void speech.prepare();stopCamera('Practice mode:
+     tap or click a message, hold with a mouse, or use Tab and
+     Enter.');mode='practice';document.body.classList.add('control-
+     active');window.scrollTo(0,0);$('mode-label').textContent='Practice mode \xb7 tap, click, or
+     hold with a mouse.';$('pause').disabled=false;resetDwell();});
136   document.querySelector('.communication').addEventListener('pointermove',event=>{if(mode==='pra
+     ctice'&&event.pointerType!=='touch')mousePoint={x:event.clientX,y:event.clientY};});
137   document.querySelector('.communication').addEventListener('pointerleave',()=>{if(mode==='pract
+     ice'){mousePoint=null;pointer.hidden=true;resetDwell();}});
138   $('pause').addEventListener('click',togglePause);$('confirm').addEventListener('click',confirm
+     );$('cancel').addEventListener('click',dismiss);$('repeat').addEventListener('click',()=>speak
+     (confirmed));$('clear').addEventListener('click',clear);
139   $('repeat').dataset.target='repeat';$('clear').dataset.target='clear';
```

dist/app.mjs (continued)

```
140   $('language').addEventListener('change',()=>{dismiss();lang=$('language').value;speech.cancel(
+     );render();voiceInfo();});
141   $('settings-button').addEventListener('click',()=>{voiceInfo();showDialog('settings-
+     dialog');});$('settings-done').addEventListener('click',()=>$('settings-dialog').close());
142   $('voice-mode').addEventListener('change',event=>{speech.cancel();speech.mode=event.target.val
+     ue;voiceInfo();});
143   $('dwell').addEventListener('input',event=>{gate.duration=Number(event.target.value)*1000;$('d
+     well-output').textContent=event.target.value+' seconds';$('hold-label').textContent='Hold to
+     select \xb7 '+event.target.value+' seconds';resetDwell();});
144   $('sensitivity').addEventListener('input',event=>{sensitivity=Number(event.target.value);$('se
+     nsitivity-output').textContent=sensitivity.toFixed(1)+'\xd7';steadyPointer.reset();point=null;
+     resetDwell(true);});
145   $('smoothing').addEventListener('change',event=>{alpha=Number(event.target.value);$('smoothing
+     -output').textContent=event.target.selectedOptions[0].textContent;resetDwell(true);});$('rever
+     se-
+     x').addEventListener('change',event=>{reverseX=event.target.checked;steadyPointer.reset();poin
+     t=null;resetDwell(true);});
146   $('edit-phrases').addEventListener('click',()=>{const fields=$('phrase-
+     fields');fields.replaceChildren();packs[lang].forEach((text,i)=>{const
+     label=document.createElement('label');label.className='phrase-
+     field';label.textContent=hints[i];const input=document.createElement('input');input.value=text
+     ;input.maxLength=160;input.lang=lang;input.dir=['ur','ar'].includes(lang)?'rtl':'ltr';label.ap
+     pend(input);fields.append(label);});$('phrase-error').textContent='';showDialog('phrases-
+     dialog');});
147   $('save-phrases').addEventListener('click',()=>{const values=[...$('phrase-
+     fields').querySelectorAll('input')].map(input=>input.value.trim());if(values.some(v=>!v)){
+     $('phrase-error').textContent='Please give every card a
+     message.';return;}packs[lang]=values;dismiss();$('phrases-dialog').close();render();});
148   dialogs.forEach(dialog=>dialog.addEventListener('close',()=>resetDwell(true)));
149   window.addEventListener('resize',()=>{steadyPointer.reset();point=null;mousePoint=null;resetDw
+     ell(true);});
150   document.addEventListener('keydown',event=>{if(event.key==='Escape'&&!dialogOpen()&&draft)dism
+     iss();if(event.code==='Space'&&!dialogOpen()&&!event.target.closest('button,input,select,texta
+     rea')&&mode!=='click'){event.preventDefault();togglePause();}});
151   document.addEventListener('visibilitychange',()=>{if(document.hidden){speech.cancel();if(mode=
+     =='camera')stopCamera('Camera stopped because this tab was hidden. Enable it again when ready.
+     ');else{mousePoint=null;resetDwell();}}});window.addEventListener('pagehide',()=>stopCamera())
+     ;
152   window.speechSynthesis?.addEventListener('voiceschanged',voiceInfo);
153   const toolContext=document.modelContext||navigator.modelContext;
154   if(toolContext?.registerTool){const lifecycle=new AbortController();try{Promise.resolve(toolCo
+     ntext.registerTool({name:'stage_careswitch_message',title:'Select a communication
+     message',description:'Select a CareSwitch communication card for review. The user must confirm
+      separately. Does not speak, start a camera, or contact anyone.',annotations:{readOnlyHint:fal
+     se,untrustedContentHint:true},inputSchema:{type:'object',properties:{messageId:{type:'string',
+     enum:ids}},required:['messageId'],additionalProperties:false},execute:async input=>{if(!input|
+     |!ids.includes(input.messageId)||Object.keys(input).some(k=>k!=='messageId'))throw new
+     Error('Provide one valid messageId.');stage(input.messageId);return{selectedMessage:draft.text
+     ,messageId:draft.id,confirmed:false};}},{signal:lifecycle.signal})).catch(()=>{});}catch{}wind
+     ow.addEventListener('pagehide',()=>lifecycle.abort(),{once:true});}
155   render();voiceInfo();requestAnimationFrame(animate);
```

dist/core.mjs

```
export const clamp=(n,min,max)=>Math.max(min,Math.min(max,n));
export function median(values){if(!values.length)return NaN;const a=[...values].sort((x,y)=>x-y),m=Math.floor(a.length/2);return a.length%2?a[m]:(a[m-1]+a[m])/2;}
export function makeCalibration(samples){
 const points={};for(const name of ['center','left','right','up','down']){const a=samples[name]||[];if(a.length<8)throw new Error('Not enough stable face samples. Try setup again in better light.');points[name]={x:median(a.map(p=>p.x)),y:median(a.map(p=>p.y))};}
 const xLow=Math.min(points.left.x,points.right.x),xHigh=Math.max(points.left.x,points.right.x),yLow=Math.min(points.up.y,points.down.y),yHigh=Math.max(points.up.y,points.down.y);
 if(xHigh-xLow<0.025||yHigh-yLow<0.018)throw new Error('The movement range was too small to distinguish reliably. Try comfortable movements again, or use practice mode.');
 if(points.center.x<=xLow||points.center.x>=xHigh||points.center.y<=yLow||points.center.y>=yHigh)throw new Error('The center position was outside the movement range. Keep the camera still and repeat setup.');
 return{center:points.center,xLow,xHigh,yLow,yHigh,xSign:points.right.x>points.left.x?1:-1,ySign:points.down.y>points.up.y?1:-1};
}
export function mapPointer(nose,calibration,sensitivity=1,reverseX=false){
 if(!nose||!Number.isFinite(nose.x)||!Number.isFinite(nose.y))return null;
 const c=calibration.center,dx=nose.x-c.x,dy=nose.y-c.y;
 const sx=dx>=0?calibration.xHigh-c.x:c.x-calibration.xLow,sy=dy>=0?calibration.yHigh-c.y:c.y-calibration.yLow;
 // Soften small movements while preserving the full comfortable calibrated range.
 const curve=(delta,span)=>Math.sign(delta)*Math.pow(Math.abs(delta)/Math.max(.005,span),1/clamp(sensitivity,.3,2));
 return{x:clamp(.5+curve(dx,sx)*.5*calibration.xSign*(reverseX?-1:1),0,1),y:clamp(.5+curve(dy,sy)*.5*calibration.ySign,0,1)};
}
export class DwellGate{
 constructor(duration=1500){this.duration=duration;this.reset();}
 reset(){this.target=null;this.started=0;this.blocked=null;this.lastTime=0;}
 update(target,now,enabled=true){
  if(!enabled){this.reset();return{progress:0,fire:null};}
  if(!target){this.target=null;this.blocked=null;this.started=now;this.lastTime=now;return{progress:0,fire:null};}
  if(now-this.lastTime>350||target!==this.target){this.target=target;this.started=now;}
  this.lastTime=now;
  if(this.blocked===target)return{progress:0,fire:null};
  const progress=clamp((now-this.started)/this.duration,0,1);
  if(progress>=1){this.blocked=target;return{progress:1,fire:target};}
  return{progress,fire:null};
 }
}
```

dist/pointer.mjs

```
import {clamp,median} from './core.mjs';

export class SteadyPointer {
 constructor({deadbandPixels=8,maxSpeedPixels=480}={}){
  this.deadbandPixels=deadbandPixels;this.maxSpeedPixels=maxSpeedPixels;this.reset();
 }
 reset(){this.history=[];this.target=null;this.position=null;this.lastTime=null;}
 push(point,{width,height}){
  if(!point||!Number.isFinite(point.x)||!Number.isFinite(point.y))return false;
  this.history.push({x:clamp(point.x,0,1),y:clamp(point.y,0,1)});
  if(this.history.length>3)this.history.shift();
  const next={x:median(this.history.map(p=>p.x)),y:median(this.history.map(p=>p.y))};
  if(!this.target){this.target=next;this.position={...next};return true;}
  const
dx=(next.x-this.target.x)*width,dy=(next.y-this.target.y)*height,distance=Math.hypot(dx,dy);
  if(distance<=this.deadbandPixels)return true;
  // Subtract the deadband instead of jumping when the threshold is crossed.
  const amount=(distance-this.deadbandPixels)/distance;
  this.target={x:this.target.x+(next.x-this.target.x)*amount,y:this.target.y+(next.y-
this.target.y)*amount};
  return true;
 }
 step(now,{width,height,alpha=.12}){
  if(!this.position||!this.target)return null;
  if(this.lastTime===null){this.lastTime=now;return{...this.position};}
  const elapsed=clamp(now-this.lastTime,0,50);this.lastTime=now;
  const dx=this.target.x-this.position.x,dy=this.target.y-
this.position.y,distance=Math.hypot(dx*width,dy*height);
  if(distance<.5){this.position={...this.target};return{...this.position};}
  // Match smoothing across display refresh rates and cap travel in pixels/sec.
  const smooth=1-Math.pow(1-clamp(alpha,.01,.95),elapsed/85);
  const amount=Math.min(smooth,this.maxSpeedPixels*elapsed/1000/distance);
  this.position={x:this.position.x+dx*amount,y:this.position.y+dy*amount};
  return{...this.position};
 }
}
```

dist/camera.mjs

```
// A playing camera can deliver frames even while its play() promise is pending.
// Wait for usable frames with a cancellation check and a bounded timeout.
export async function
waitForCameraFrames(video,{isCurrent=()=>true,timeoutMs=10000,pollMs=50}={}){
 let playbackError=null;
 try{video.play()?.catch(()=>{playbackError=new Error('Camera connected, but video playback
was blocked. Try opening the demo in Chrome or Edge.');});}
 catch{throw new Error('This browser could not start the camera video. Try Chrome or Edge.');}
 const started=performance.now();
 await new Promise((resolve,reject)=>{
  function check(){
   if(!isCurrent())return reject(new Error('Camera startup canceled.'));
   if(playbackError)return reject(playbackError);
   if(video.readyState>=2&&!video.paused&&video.videoWidth>0&&video.videoHeight>0)return
resolve();
   if(performance.now()-started>=timeoutMs)return reject(new Error('Camera connected, but no
video frames arrived. Turn it off and try again, or open the demo in Chrome or Edge.'));
   setTimeout(check,pollMs);
  }
  check();
 });
}
```

dist/vision-worker.js

```
/* Classic worker: MediaPipe loads its WASM loader with importScripts. */
importScripts('./assets/mediapipe/vision_bundle.js');
let model;
self.onmessage=async ({data})=>{
 try{
  if(data.kind==='init'){
   const wasm=await Vision.FilesetResolver.forVisionTasks(new URL('./assets/mediapipe/wasm',self.location.href).href);
   model=await Vision.FaceLandmarker.createFromOptions(wasm,{baseOptions:{modelAssetPath:new URL('./assets/face_landmarker.task',self.location.href).href,delegate:'CPU'},runningMode:'VIDEO',numFaces:2,minFaceDetectionConfidence:.6,minFacePresenceConfidence:.6,minTrackingConfidence:.6,outputFaceBlendshapes:false,outputFacialTransformationMatrixes:false});
   self.postMessage({kind:'ready'});
  }else if(data.kind==='frame'){
   try{
    if(!model)throw new Error('The face model is not ready.');
    const result=model.detectForVideo(data.bitmap,data.timestamp);
    const faces=result.faceLandmarks||[],nose=faces.length===1?faces[0][1]:null;
    self.postMessage({kind:'tracking',count:faces.length,nose:nose?{x:nose.x,y:nose.y}:null,timestamp:data.timestamp,landmarkCount:faces[0]?.length||0});
   }finally{data.bitmap?.close();}
  }
 }catch(error){self.postMessage({kind:'error',message:error.message||'Face tracking could not start.'});}
};
```

dist/speech.mjs

```
export const speechLanguages={en:'English',ur:'Urdu',ar:'Arabic',hi:'Hindi',es:'Spanish',fr:'French',de:'German',zh:'Chinese (Mandarin)'};
export class SpeechPlayer{
 constructor({onStatus=()=>{},synthesis=globalThis.speechSynthesis,Utterance=globalThis.SpeechSynthesisUtterance,
  contextFactory=()=>new (globalThis.AudioContext||globalThis.webkitAudioContext)(),
  workerFactory=()=>new Worker(new URL('./speech-worker.mjs',import.meta.url),{type:'module'}),timeoutMs=60000}={}){
  Object.assign(this,{onStatus,synthesis,Utterance,contextFactory,workerFactory,timeoutMs});this.mode='auto';this.epoch=0;this.serial=0;this.pending=new Map();this.cache=new Map();
 }
 nativeVoice(lang){return this.synthesis?.getVoices().find(v=>v.localService&&v.lang.toLowerCase().split(/[-_]/)[0]===lang)||null;}
 info(lang){return this.mode==='auto'&&this.nativeVoice(lang)?'device':'basic';}
 prepare(){
  try{this.context??=this.contextFactory();if(this.context.state!=='suspended')return Promise.resolve();return new Promise(resolve=>{const timer=setTimeout(resolve,1500);this.context.resume().catch(()=>{}).finally(()=>{clearTimeout(timer);resolve();});});}catch{return Promise.resolve();}
 }
 cancel(){
  this.epoch++;this.synthesis?.cancel();if(this.source){this.source.onended=null;try{this.source.stop();}catch{}this.source.disconnect();this.source=null;}
 }
 request(message){
  if(!this.worker){this.worker=this.workerFactory();this.worker.onmessage=({data})=>{const p=this.pending.get(data.id);if(!p)return;clearTimeout(p.timer);this.pending.delete(data.id);data.kind==='error'?p.reject(new Error(data.message)):p.resolve(data);};
   this.worker.onerror=()=>this.resetWorker();}
  return new Promise((resolve,reject)=>{const id=++this.serial;const timer=setTimeout(()=>this.resetWorker(),this.timeoutMs);this.pending.set(id,{resolve,reject,timer});this.worker.postMessage({...message,id});});
 }
 resetWorker(){this.worker?.terminate();this.worker=null;for(const p of this.pending.values()){clearTimeout(p.timer);p.reject(new Error('Speech loading failed or timed out. Try Read again.'));}this.pending.clear();}
 async basic(message,epoch){
  if(!speechLanguages[message.lang])throw new Error('This message language has no built-in voice.');
  await this.prepare();if(epoch!==this.epoch)return;
  if(!this.context)throw new Error('Audio is unavailable in this browser.');
  const key=message.lang+'\0'+message.text;let buffer=this.cache.get(key);
  if(!buffer){this.onStatus('Loading the free '+speechLanguages[message.lang]+' voice on this device\u2026');const result=await this.request({kind:'speak',lang:message.lang,text:message.text});if(epoch!==this.epoch)return;
   buffer=await this.context.decodeAudioData(result.wav);if(epoch!==this.epoch)return;if(this.cache.size>=8)this.cache.delete(this.cache.keys().next().value);this.cache.set(key,buffer);}
  if(epoch!==this.epoch)return;await this.prepare();if(epoch!==this.epoch)return;
  if(this.context.state!=='running')throw new Error('Tap Read again to allow audio playback.');
  const source=this.context.createBufferSource();source.buffer=buffer;source.connect(this.context.destination);this.source=source;
  source.onended=()=>{source.disconnect();if(epoch!==this.epoch)return;this.source=null;this.onStatus('Message read with the free built-in voice.');};
  this.onStatus('Reading with the free '+speechLanguages[message.lang]+' voice \xb7 on this device.');source.start();
 }
 speak(message){
  if(!message)return;this.cancel();const epoch=this.epoch;void this.prepare();
  const fail=error=>{if(epoch===this.epoch)this.onStatus('Message displayed. '+(error?.message||'Voice unavailable. Try Read again.'));};
  const voice=this.mode==='auto'?this.nativeVoice(message.lang):null;
  if(voice&&this.Utterance){try{
   const utterance=new this.Utterance(message.text);utterance.voice=voice;utterance.lang=voice.lang;
   utterance.onstart=()=>{if(epoch===this.epoch)this.onStatus('Reading with your device\u2019s local voice.');};
   utterance.onend=()=>{if(epoch===this.epoch)this.onStatus('Message confirmed. Read again if needed.');};
   utterance.onerror=()=>{if(epoch===this.epoch)this.basic(message,epoch).catch(fail);};
   this.onStatus('Message confirmed. Preparing the local voice\u2026');this.synthesis.speak(utterance);return;
  }catch{/* A failed device voice can use the built-in voice. */}}
  this.basic(message,epoch).catch(fail);
 }
```

dist/speech.mjs (continued)

```
48      }
```

dist/speech-worker.mjs

```
// Local synthesis only. eSpeak NG and its voice data are served by this Site.
const voices={en:'en',ur:'ur',ar:'ar',hi:'hi',es:'es',fr:'fr',de:'de',zh:'cmn'};
let loading;
async function engine(){
 if(!loading)loading=(async()=>{
  const [{default:factory},response]=await Promise.all([import('./assets/speech/espeak-ng.js'),fetch(new URL('./assets/speech/espeak-ng.wasm',import.meta.url))]);
  if(!response.ok)throw new Error('Speech files could not load.');
  return{factory,compiled:await WebAssembly.compile(await response.arrayBuffer())};
 })().catch(error=>{loading=null;throw error;});
 return loading;
}
let queue=Promise.resolve();
self.onmessage=({data})=>{
 queue=queue.then(async()=>{
  try{
   if(data.kind==='init'){await engine();self.postMessage({id:data.id,kind:'ready'});return;}
   if(data.kind!=='speak'||!voices[data.lang]||typeof data.text!=='string'||!data.text.trim()||data.text.length>160)throw new Error('Unsupported speech request.');
   const {factory,compiled}=await engine();
   const instance=await factory({arguments:['-v',voices[data.lang],'-s','155','-f','input.txt','-w','message.wav'],
    instantiateWasm(imports,ready){WebAssembly.instantiate(compiled,imports).then(value=>ready(value,compiled));return{};},

preRun:[module=>module.FS.writeFile('input.txt',data.text)],print:()=>{},printErr:()=>{}});
   const wav=instance.FS.readFile('message.wav').slice();if(wav.length<=44)throw new Error('No speech audio was generated.');
   self.postMessage({id:data.id,kind:'audio',voice:voices[data.lang],wav:wav.buffer},[wav.buffer]);
  }catch{self.postMessage({id:data.id,kind:'error',message:'The free voice could not load or generate audio. Try Read again.'});}
 }).catch(()=>{});
};
```

tests/core-test.mjs

SHA-256: 9c96d9cafa6db9b4b29121f73b370b97dcb6ba30e18ff004917cc0f29cbab062

```
import assert from 'node:assert/strict';
import {makeCalibration,mapPointer,DwellGate} from '../dist/core.mjs';
const samples=Object.fromEntries(Object.entries({center:[.5,.5],left:[.6,.5],right:[.4,.5],up:
[.5,.4],down:[.5,.6]}).map(([key,[x,y]])=>[key,Array.from({length:20},()=>({x,y}))]));
const calibration=makeCalibration(samples);
assert.deepEqual(mapPointer({x:.5,y:.5},calibration),{x:.5,y:.5});
assert.ok(mapPointer({x:.6,y:.5},calibration).x<.001,'mirrored head movement must still map
left');
assert.ok(mapPointer({x:.4,y:.5},calibration).x>.999);
assert.ok(mapPointer({x:.5,y:.4},calibration).y<.001);
assert.equal(mapPointer({x:NaN,y:0},calibration),null);
assert.throws(()=>makeCalibration({...samples,left:samples.center,right:samples.center}),/too
small/);
assert.throws(()=>makeCalibration({...samples,up:[]}),/Not enough/);
const gate=new DwellGate(1500);let fired=[];
for(let t=100;t<=3100;t+=100){const r=gate.update('yes',t);if(r.fire)fired.push(r.fire);}
assert.deepEqual(fired,['yes'],'stationary pointer must select once');
gate.update(null,3200);assert.equal(gate.update('yes',3300).fire,null);
gate.reset();for(let t=100;t<=1100;t+=100)gate.update('confirm',t);
assert.equal(gate.update('confirm',2000).progress,0,'tracking gap resets accumulated dwell');
gate.update('confirm',2100,false);assert.equal(gate.update('confirm',2200).progress,0,'loss of
 tracking stops selection');
gate.reset();gate.update('yes',100);gate.update('yes',300);assert.equal(gate.update('no',400).
progress,0,'changing targets starts a fresh dwell');
console.log('PASS: calibration directions/range, invalid input, duplicate selection, tracking
loss, stale frames, target changes.');
```

tests/pointer-test.mjs

SHA-256: d833e6c2d17642f6ba92c7185454483acdaa128751c95a1ba5eb83177820f2d2

```
import assert from 'node:assert/strict';
import {SteadyPointer} from '../dist/pointer.mjs';
import {makeCalibration,mapPointer} from '../dist/core.mjs';
const area={width:1000,height:500,alpha:.12};
const pixels=(a,b)=>Math.hypot((a.x-b.x)*area.width,(a.y-b.y)*area.height);
const filter=new SteadyPointer();
for(let i=0;i<3;i++)filter.push({x:.5,y:.5},area);
filter.step(0,area);
for(let t=16;t<2000;t+=16){if(t%80===0)filter.push({x:.5+(t%160===0?.004:-
.004),y:.5+.004},area);assert.deepEqual(filter.step(t,area),{x:.5,y:.5},'small stationary
noise must not move the pointer');}
filter.push({x:1,y:0},area);assert.deepEqual(filter.step(2016,area),{x:.5,y:.5},'one landmark
spike must not jump the pointer');
filter.push({x:.5,y:.5},area);filter.push({x:.5,y:.5},area);
let previous=filter.step(2032,area);
for(let t=2048;t<7048;t+=16){if(t%80===0)filter.push({x:.9,y:.9},area);const
next=filter.step(t,area);assert.ok(pixels(next,previous)<=480*.016+.001,'each animation step
must respect speed limit');previous=next;}
assert.ok(previous.x>.87&&previous.y>.87,'deliberate movement must remain reachable');
const beforeGap=previous;const
afterGap=filter.step(20000,area);assert.ok(pixels(afterGap,beforeGap)<=480*.05+.001,'long
frame gaps must not teleport the pointer');
filter.reset();assert.equal(filter.step(21000,area),null,'tracking reset must discard old
movement');
assert.equal(filter.push({x:NaN,y:1},area),false);
function response(frameMs){const f=new
SteadyPointer();f.push({x:.5,y:.5},area);f.step(0,area);for(let
i=0;i<3;i++)f.push({x:.85,y:.8},area);let output;for(let
t=frameMs;t<=2000;t+=frameMs)output=f.step(t,area);return output;}
assert.ok(pixels(response(1000/30),response(1000/60))<5,'smoothing must be consistent across
refresh rates');
const samples=Object.fromEntries(Object.entries({center:[.5,.5],left:[.6,.5],right:[.4,.5],up:
[.5,.4],down:[.5,.6]}).map(([key,[x,y]])=>[key,Array.from({length:20},()=>({x,y}))]));
const calibration=makeCalibration(samples);
assert.ok(mapPointer({x:.42,y:.5},calibration,.8).x<mapPointer({x:.42,y:.5},calibration,1).x,'
lower sensitivity must reduce displacement');
for(const [nose,side] of [[{x:.6,y:.5},'left'],[{x:.4,y:.5},'right']]){const
p=mapPointer(nose,calibration,.8);assert.ok(side==='left'?p.x<.15:p.x>.85,'default sensitivity
 must still reach outer message columns');}
assert.equal(mapPointer({x:.6,y:.5},calibration,.6).x,0,'even low sensitivity must preserve
calibrated left edge');
assert.equal(mapPointer({x:.4,y:.5},calibration,.6).x,1,'even low sensitivity must preserve
calibrated right edge');
assert.equal(mapPointer({x:.5,y:.4},calibration,.6).y,0,'pause at the top must remain
reachable');
assert.equal(mapPointer({x:.5,y:.6},calibration,.6).y,1,'bottom controls must remain
reachable');
console.log('PASS: small jitter held, spike rejected, speed bounded, deliberate travel, stale
reset, 30/60 Hz consistency, and lower sensitivity.');
```

tests/camera-test.mjs

SHA-256: 5f07924adfde5c44a4b87dfec69ed50070c931a96f2044cdf12335c980123587

```
1     import assert from 'node:assert/strict';
2     import {waitForCameraFrames} from '../dist/camera.mjs';
3     const ready={readyState:4,paused:false,videoWidth:640,videoHeight:480,play:()=>new
+     Promise(()=>{})};
4     await waitForCameraFrames(ready,{timeoutMs:100,pollMs:5});
5     const delayed={...ready,readyState:0};setTimeout(()=>{delayed.readyState=4;},20);await
+     waitForCameraFrames(delayed,{timeoutMs:100,pollMs:5});
6     await assert.rejects(waitForCameraFrames({...ready,readyState:0},{timeoutMs:20,pollMs:5}),/no
+     video frames/);
7     await
+     assert.rejects(waitForCameraFrames({...ready,readyState:0},{isCurrent:()=>false}),/canceled/);
8     await assert.rejects(waitForCameraFrames({...ready,readyState:0,play:()=>Promise.reject(new
+     Error('blocked'))},{timeoutMs:100,pollMs:5}),/playback was blocked/);
9     console.log('PASS: pending play promise with real frames, delayed frames, timeout,
+     cancellation, and playback rejection.');
```

tests/speech-test.mjs

SHA-256: 181f22e2b9086502699f52b84543a7f7981834d61ec7f7a5104039dc288e0230

```
1     import assert from 'node:assert/strict';import {SpeechPlayer} from '../dist/speech.mjs';
2     const flush=async()=>{for(let i=0;i<12;i++)await Promise.resolve();};
3     let starts=0,stops=0,requests=[],statuses=[],utterances=[];
4     const worker={postMessage:m=>requests.push(m),terminate:()=>{}};
5     const context={state:'running',destination:{},decodeAudioData:async()=>({}),createBufferSource
+     :()=>({connect(){},disconnect(){},start(){starts++;},stop(){stops++;}})};
6     const native={getVoices:()=>[{lang:'en-US',localService:true},{lang:'ur-
+     PK',localService:false}],cancel(){},speak:u=>utterances.push(u)};
7     const p=new SpeechPlayer({onStatus:s=>statuses.push(s),synthesis:native,Utterance:class{constr
+     uctor(t){this.text=t;}},contextFactory:()=>context,workerFactory:()=>worker,timeoutMs:2000});
8     assert.equal(p.nativeVoice('ur'),null);p.speak({lang:'ur',text:'\u0645\u062c\u06be\u06d2
+     \u0648\u0642\u0641\u06c1 \u0686\u0627\u06c1\u06cc\u06d2\u06d4'});await
+     flush();assert.equal(requests.length,1);const
+     first=requests[0];p.cancel();worker.onmessage({data:{id:first.id,kind:'audio',wav:new
+     ArrayBuffer(100)}});await flush();assert.equal(starts,0,'A canceled message must never play
+     after synthesis finishes');
9     p.speak({lang:'ur',text:'\u0645\u062c\u06be\u06d2 \u0648\u0642\u0641\u06c1
+     \u0686\u0627\u06c1\u06cc\u06d2\u06d4'});await
+     flush();worker.onmessage({data:{id:requests[1].id,kind:'audio',wav:new
+     ArrayBuffer(100)}});await flush();assert.equal(starts,1);p.cancel();assert.equal(stops,1);
10    p.speak({lang:'en',text:'Yes.'});assert.equal(utterances.length,1);p.cancel();const before=sta
+     tuses.length;utterances[0].onerror();utterances[0].onend();assert.equal(statuses.length,before
+     ,'Canceled native events must not change the current status');
11    p.mode='basic';p.speak({lang:'en',text:'No.'});await
+     flush();assert.equal(requests.at(-1).lang,'en');p.resetWorker();await
+     flush();assert.match(statuses.at(-1),/Message displayed/);
12    let blocked=[];const q=new SpeechPlayer({onStatus:s=>blocked.push(s),synthesis:null,contextFac
+     tory:()=>({...context,state:'suspended',resume:async()=>{}}),workerFactory:()=>worker});q.cach
+     e.set('ur\0\u0646\u06c1\u06cc\u06ba\u06d4',{});q.speak({lang:'ur',text:'\u0646\u06c1\u06cc\u06
+     ba\u06d4'});await flush();assert.match(blocked.at(-1),/Tap Read
+     again/);assert.equal(starts,1);
13    console.log('PASS: local-only voice selection, stale audio cancellation, stop playback,
+     canceled native callbacks, forced free voice, worker failure, and blocked audio fallback.');
```